# Matrices

Lecture and worked examples

# Objectives

Compute with matrices (sum, product by a real number, product); find the inverse of a matrix of order 2, and of order 3 with a calculator; compute a power, in particular by induction; solve a linear system in matrix form; represent a transformation of the plane by a matrix.

# Introductory activity

We represent a point $M(x, y)$ by the column $X = \begin{pmatrix} x \\ y \end{pmatrix}$. For a matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we define the product $AX = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}$: it is the column of a new point.

# Introductory activity (2)

1. Let $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Compute the images under $R$ of the vertices of the triangle $O(0,0)$, $P(2,0)$, $Q(2,1)$. Draw both triangles. Which transformation do you recognise?
2. Let $E = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$. Same question.
3. Compute $R(EX)$ for any $X = \begin{pmatrix} x \\ y \end{pmatrix}$. Find a matrix $C$ such that $R(EX) = CX$ for every $X$.
4. Compute likewise $E(RX)$ and the corresponding matrix. Are the two composite transformations the same? Check on the point $Q$.

# Introductory activity (3)

5. Suggest a rule for computing $C$ directly from $R$ and $E$: it is the matrix product $RE$.

# Introductory activity (4)

*Observing is not proving: the properties conjectured here are established in the course.*

# Vocabulary and operations

A *matrix* of size $n \times p$ is an array of real numbers with $n$ rows and $p$ columns. The entry in row $i$ and column $j$ of $A$ is written $a_{ij}$. An $n \times 1$ matrix is a *column* matrix, a $1 \times p$ matrix a *row* matrix, an $n \times n$ matrix a *square* matrix of order $n$. The *identity* matrix $I_n$ has 1s on the diagonal and 0s elsewhere; the *zero* matrix has only 0s. A square matrix is *diagonal* if its entries off the diagonal are zero.

# Vocabulary and operations

For $A$ and $B$ of the same size and $k$ real, $A + B$ and $kA$ are computed entry by entry.

If $A$ has size $n \times p$ and $B$ has size $p \times q$, the product $AB$ is the $n \times q$ matrix with entries

$$c_{ij} = \sum_{k=1}^{p} a_{ik} b_{kj} = a_{i1} b_{1j} + a_{i2} b_{2j} + \cdots + a_{ip} b_{pj}.$$

# Vocabulary and operations

$$\begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 4 & -2 \end{pmatrix} = \begin{pmatrix} 1 \times 0 + 2 \times 4 & 1 \times 1 + 2 \times (-2) \\ -1 \times 0 + 3 \times 4 & -1 \times 1 + 3 \times (-2) \end{pmatrix} = \begin{pmatrix} 8 & -3 \\ 12 & -7 \end{pmatrix}.$$

# Vocabulary and operations

When the products are defined: $(AB)C = A(BC)$, $A(B+C) = AB+AC$, $(A+B)C = AC+BC$, $(kA)B = k(AB)$, and $AI_n = I_nA = A$ for $A$ square of order $n$.

# Vocabulary and operations

1. In general $AB \neq BA$; consequently $(A+B)^2 = A^2 + AB + BA + B^2$, and not $A^2 + 2AB + B^2$.
2. A product can be zero without either factor being zero: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

# Powers

For $A$ square of order $n$, let $A^0 = I_n$ and $A^{k+1} = A^k A$ for every natural number $k$.

# Powers

For $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, we compute $A^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, $A^3 = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$ and conjecture $A^k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$.

This is true for $k = 0$; if it is true for $k$, then $A^{k+1} = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & k+1 \\ 0 & 1 \end{pmatrix}$.

# Powers

If $D$ is diagonal with diagonal entries $d_1, \ldots, d_n$, then $D^k$ is diagonal with entries $d_1^k, \ldots, d_n^k$.

# Inverse and systems

A square matrix $A$ of order $n$ is *invertible* if there is a matrix $B$ such that $AB = BA = I_n$. This matrix is unique; it is written $A^{-1}$.

# Inverse and systems

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The matrix $A$ is invertible if and only if $ad - bc \neq 0$, and then

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The number $ad - bc$ is the *determinant* of $A$, written $\det A$.

# Inverse and systems

Let $M = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. Computation gives $AM = MA = (ad - bc)I_2$. If $ad - bc \neq 0$, the matrix $\frac{1}{ad - bc} M$ is therefore the inverse of $A$.

If $ad - bc = 0$, then $AM = 0$. Suppose $A$ is invertible: multiplying on the left by $A^{-1}$ gives $M = 0$, so $a = b = c = d = 0$ and $A$ is zero. But the zero matrix is not invertible ($0 \times B = 0 \neq I_2$): a contradiction.

# Inverse and systems

A linear system of $n$ equations in $n$ unknowns can be written $AX = B$, where $A$ is the matrix of coefficients, $X$ the column of unknowns and $B$ the column of right-hand sides. If $A$ is invertible, the system has a unique solution: $X = A^{-1}B$.

# Inverse and systems

If $AX = B$, then $A^{-1}AX = A^{-1}B$, that is $X = A^{-1}B$. Conversely, $A\left(A^{-1}B\right) = B$.

# Inverse and systems

For a matrix of order 3, obtain $A^{-1}$ with a calculator or software, then check one or two entries of $AA^{-1}$ by hand.

The rotation with centre $O$ and angle $\theta$ maps the point with column $X$ to the point with column $R_\theta X$, where

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

# Transformations of the plane and sequences

The point $M(x, y)$ has affix $z = x + iy$, and its image has affix $e^{i\theta}z = (\cos\theta + i\sin\theta)(x + iy) = (x\cos\theta - y\sin\theta) + i(x\sin\theta + y\cos\theta)$ (chapter 3): these are the entries of $R_\theta X$.

# Transformations of the plane and sequences

If $X_{k+1} = AX_k$ for every $k$, then $X_k = A^k X_0$ (immediate induction). A sequence defined by $u_{k+2} = au_{k+1} + bu_k$ reduces to this with $X_k = \begin{pmatrix} u_{k+1} \\ u_k \end{pmatrix}$ and $A = \begin{pmatrix} a & b \\ 1 & 0 \end{pmatrix}$.